INCH-POUND

MS25459J
10 December 2020
SUPERSEDING
MS25459H
27 November 2003

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 5 AMPERES, 4 PDT, TYPE I, MAGNETIC LATCH, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 5 JUNE 1987.
NO SUPERSEDING SPECIFICATION. (FOR NEW DESIGN USE MIL-PRF-83536/18, MIL-PRF-83536/19, MIL-PRF-83536/20)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

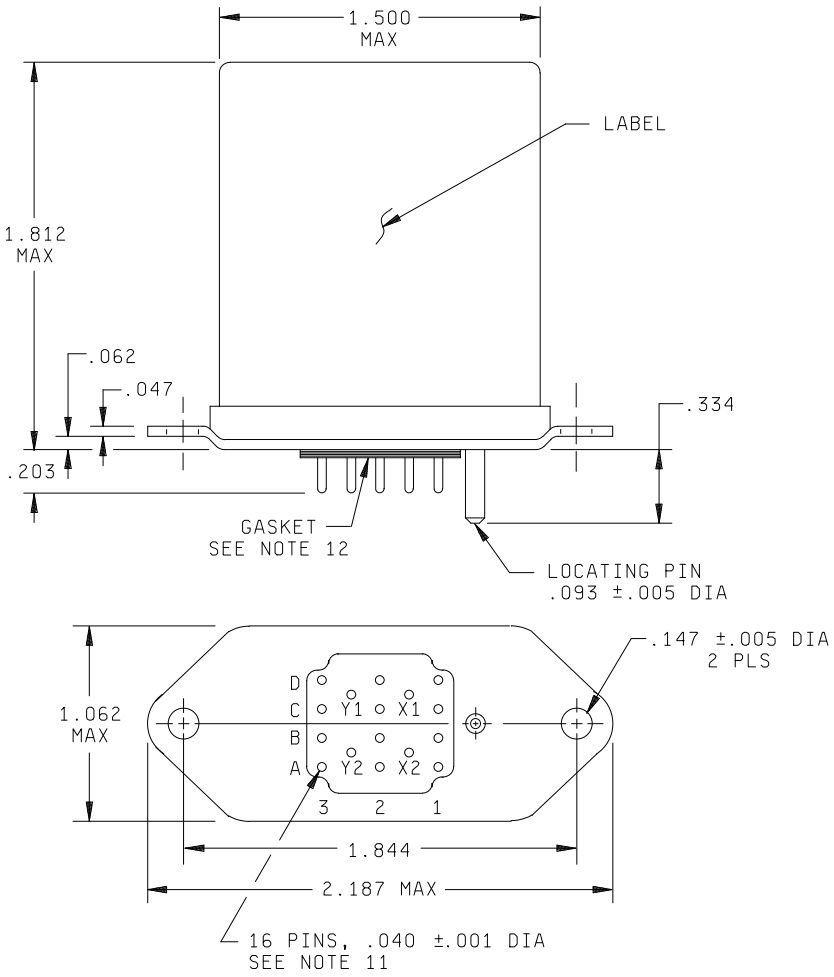
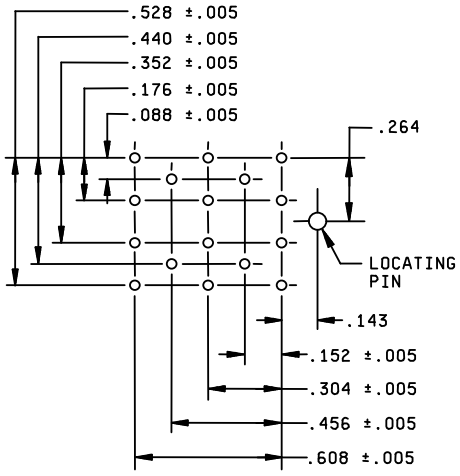


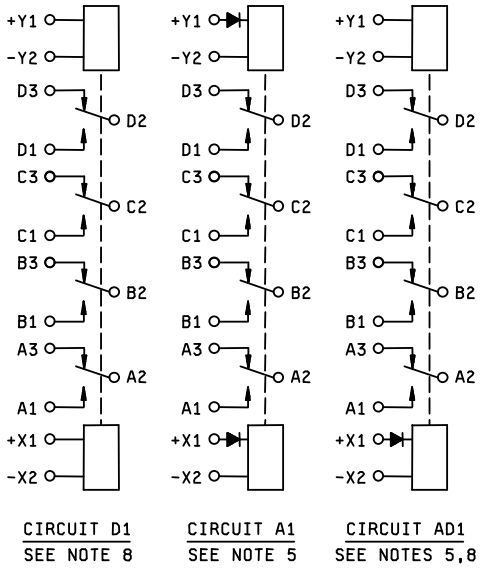
FIGURE 1. Dimensions and configurations.

AMSC N/A

FSC 5945

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Inches	mm	Inches	mm
.001	0.03	.264	6.71
.005	0.13	.304	7.72
.040	1.02	.334	8.48
.047	1.19	.352	8.94
.062	1.57	.440	11.18
.075	1.91	.456	11.58
.088	2.24	.528	13.41
.093	2.36	.608	15.44
.143	3.63	1.062	26.97
.147	3.73	1.500	38.10
.152	3.86	1.812	46.02
.176	4.47	1.844	46.84
.203	5.16	2.187	55.55

TABLE I. Dash numbers and characteristics.

Dash number MS25459-	Type	Coil	Terminal type	Max weight in pounds
D1	I	dc	Plug in	0.30
A1	I	ac	Plug in	0.32
AD1	I	ac-dc	Plug in	0.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified hereon.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Pins to be perpendicular to header surface within 1 degree.
7. Shock, vibration, and acceleration requirements application with coils de-energized.
8. Relay is magnetically latched in both positions. Caution note to observe polarity must appear on relays with dc coils.
9. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
10. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this standard to the extent specified herein.

FIGURE 1. Dimensions and configurations. - Continued.

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11. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals shall be gold plated. One system for gold plating that may be used is ASTM-B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
12. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have thickness .075, recessed .047 in to bracket. Gasket material according to SAE-AMS3332 has been considered acceptable.

FIGURE 1. Dimensions and configurations. - Continued.

TABLE II. Operating characteristics.

PIN MS25459-	Coil data											Time - milliseconds max				
	Coil	Rated			Max		Max pick-up voltage			Drop out voltage	Operate 3/	Release 4/	Contact Bounce			
													Main		Aux	
		Volts 1/	Freq Hz	Res Ω	Volts	Amp	Normal 2/	High temp test	Cont current test				NO	NC	NO	NC
D1	X1, X2 Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2	N/A	N/A
A1	X1, X2 Y1, Y2	115	400 5/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2	N/A	N/A
AD1	X1, X2	115	400 5/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2	N/A	N/A
	Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2	N/A	N/A

1/ CAUTION: Use of any coil voltage less than rated voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.
5/ MS25459-A1 and AD1 ac coils may be used on 60 Hz if maximum ambient temperature is limited to +85°C (coil current 0.066 ampere maximum).

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	5	5			5	4							
Inductive	100													
Inductive	20	3	3			3	2							
Motor	100	1.5	1.5			1.5	1							
Lamp	100	0.8	0.8			0.8	0.6							
Transfer load														2/
Mechanical life reduced current	400	1.25	1.25			1.25	1							
Mixed loads	Applicable per specification MIL-PRF-6106													

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Environmental characteristics:

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 g's
Duration	11 ms
Max duration contact opening	10 μs
Vibration - sinusoidal	
G-level	10 g's
Frequency range	5 - 1,500 Hz
Acceleration	15 g's

Electrical characteristics:

Insulation resistance:

Initial: 100 megohms.

After life or environmental tests: 50 megohms.

Dielectric strength (sea level):

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,000 V rms	1,000 V rms

Dielectric strength (altitude):

	(When mounted in mating socket) 80,000 ft
Coil to case	500 V rms
Aux contacts	
All other points	500 V rms

Max contact drop initial	0.150 volt.
After life test	0.175 volt.
Overload current	40 amperes dc, 60 amperes ac.
Rupture current	50 amperes dc, 80 amperes ac.
Duty rating	Continuous.
RFI specification	MIL-STD-461. (Applicable to coil circuits of ac operated relays.)

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Conformance inspection.

Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military specification sheet.

Performance of groups B and C tests are not applicable.

Qualification by similarity: See MIL-PRF-6106.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461 SAE-AMS3332 ASTM-B488

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2020-056)

Review activity:
Navy - EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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